

Michal Matusik

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EDUCATION

Purdue University | West Lafayette, IN

Bachelor of Science in Computer Science | Expected May 2030

TECHNICAL SKILLS

Languages: Python - advanced SQL - proficient, C++ - basic familiarity JavaScript - currently learning, **ML & Data:** PyTorch, scikit-learn, XGBoost, NLTK, HG Transformers, ONNX Runtime, NumPy, pandas, OpenCV, PyArrow, SQLite, asyncio

Engineering: Git, Firebase, Supabase, WebSockets, REST APIs, Linux, AWS EC2/S3

AI-assisted development: Claude Code, MCPs, hooks, plugins

SELECTED PROJECTS

FitPerfect - On-Device Fitness Form Coach | *Python, Flutter, ONNX Runtime* | **Apr. 2024 - May 2025**

Built a private-by-default iOS MVP that turns exercise video into local form feedback, rep counts, joint-angle analysis, and 3D motion data for squats and overhead presses.

Hand-built the ML pipeline: YOLOv8 person detection, RTMPose keypoints, COCO-to-H36M conversion, and MotionBERT temporal 3D pose lifting; used AI-assisted development for Flutter mobile implementation.

Engineered coordinate transforms, camera calibration, and ONNX inference to keep skeleton overlays aligned under real capture conditions.

Multi-Task Denoising & Noise Characterization | *Python, PyTorch, U-Net* | [GitHub](#)

Solved an AI Olympiad multi-task vision problem with a compact U-Net that jointly denoises 28x28 images and classifies Gaussian vs. uniform noise using a combined MSE + binary cross-entropy objective.

Achieved 29.28 dB validation PSNR, 100% classification accuracy, and 0.00136 / 0.00120 MSE for Gaussian μ/σ estimation, earning 100/100; recovered parameters analytically from denoising residuals without a regression head.

Chess Quant - Live Chess Tilt Detection Prototype | *Next.js, TypeScript, Python, XGBoost, ONNX* | **May 2026 - Jun. 2026**

Built an applied-ML product that analyzes recent Lichess games and estimates when continued play may have negative expected value, combining session-level accuracy, blunder, pace, streak, and break signals.

Trained an XGBoost model on 1M GPU-processed games from a 90M-game corpus; retrospective simulations using the selected stopping threshold avoided 56 losses on average.

Implemented a full-stack workflow with Lichess OAuth PKCE, Stockfish in a Web Worker, ONNX-backed inference, Firebase.

Scientific Claim Retrieval on SciFact | *Python, PyTorch, Transformers, SGPT* | [GitHub](#)

Built an embedding-based scientific-source retriever using SGPT-125M weighted-mean pooling and asymmetric query/document tokens; ranks 5,183 SciFact papers for 300 claims by cosine similarity.

Achieved nDCG@10 of 0.551 (100/100 validation score) under 768-dimensional embedding and offline 10-minute evaluation constraints; encoded the full corpus and query set in about 90 seconds on CPU/MPS.

ACHIEVEMENT

AI Olympiad National Finalist - advanced to the national final round through competitive AI and machine-learning problem solving.